JIAXI NIE

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Research interests

My research interests lie mainly in extremal combinatorics and probabilistic combinatorics, but I keep my brain open for any interesting math problems.

Positions

Visiting Assistant Professor, Georgia Institute of Technology
Supervisor: Xiaoyu He 2024 August – present

Postdoctoral Researcher, Shanghai Center for Mathematical Sciences
Supervisor: Hehui Wu 2022 September – 2024 July

Education

Ph.D. in Mathematics, University of California, San Diego 2016 – 2022
Advisor: Jacques Verstraëte. *GPA: 4.00/4.00*

Exchange Semester, University of California, Berkeley 2014 Fall
GPA: 4.00/4.00

B.S. in Mathematics, Nankai University 2012 – 2016
GPA: 90.41/100

Papers

Ordered by submission time. * for selected.

30. *Block Designs with Large Transversal-Block Intersections (Negative answer to an Erdős problem)
Submitted. arxiv.

29. Matching and intersection problems for non-trivial r-partite r-uniform hypergraphs (with P. Frankl)
Submitted. arxiv.

28. *Hypergraph Ramsey numbers with quasipolynomial growth rate (with X. He, L. Post, J. Verstraëte) *Submitted. arxiv.*

27. New Sidorenko-type inequalities in tournaments (With X. He, N. Mani, N. Tung and F. Wei)
Submitted. arxiv.

26. Decomposition of Cliques into k-Star-Forests (With Y. Ren and H. Wu)
Submitted. arxiv.

25. *Maximum in-general-position set in a random subset of $\mathbb{F}_q^d$ (With Y. Chen, J. Yu and W. Zhang)
Journal of Combinatorial Theory, Series B (2026). arxiv.

24. *Evasive sets, twisted varieties, and container-clique trees (With J. Lim and J. Zeng)
SODA 2026. arxiv.

23. *Off-Diagonal Ramsey Numbers for Linear Hypergraphs (With X. He, Y. Wigderson and H. Yu)
Combinatorics, Probability and Computing (2026). arxiv.

22. Supersaturation of odd linear cycles (With L. Deng, J. Han and S. Spiro)
Electronic Journal of Combinatorics (2026). arxiv.

21. Generalized Erdős-Rogers problems for hypergraphs (With X. He)
European Journal of Combinatorics (2026). arxiv.

20. On tight tree-complete hypergraph Ramsey numbers
Journal of Graph Theory (2026). arxiv.

19. Random Turán Problems for $K_{s,t}$ Expansions (With S. Spiro)
To appear on SIAM Journal on Discrete Mathematics. arxiv.

18. On the Matching Problem in Random Hypergraphs (With P. Frankl and J. Wang)
Discrete Mathematics (2026). arxiv.

17. *Off-diagonal Ramsey number for slowly growing hypergraphs (with S. Mattheus, D. Mubayi and J. Verstraëte)
Random Structures & Algorithms (2025). arxiv.

16. On odd covers of cliques and disjoint unions. (with C. Buchanan, A. Clifton, E. Culver, P. Frankl, K. Ozeki, P. Rombach, and M. Yin)
To appear on Journal of Graph Theory. arxiv.

15. *Random Turán Problems for Hypergraph Expansions. (With S. Spiro)
Submitted. arxiv.

14. *Sidorenko Hypergraphs and Random Turán Numbers. (With S. Spiro)
Combinatorial Theory (2025). arxiv.

13. *Random Turán and counting results for general position sets over finite fields. (with Y. Chen, X. Liu, and J. Zeng)
SCIENCE CHINA Mathematics (2025). arxiv.

12. Random Turán theorem for expansions of spanning subgraphs of tight trees.
SIAM Journal on Discrete Mathematics (2026). arxiv.

11. *Turán theorems for even cycles in random hypergraph.
Journal of Combinatorial Theory, Series B (2024). arxiv.

10. The Number of Edge Colorings with Small Independence Number and No Monochromatic H. (with X. Hu and Q. Lin)
The Electronic Journal of Combinatorics (2025). arxiv.

9. On asymptotic local Turán problems. (with P. Frankl)
The Moscow Journal of Combinatorics and Number Theory (2023). arxiv.

8. On asymptotic packing of convex geometric and ordered graphs. (with E. Surya, and J. Zeng)
Journal of Graph Theory (2023). arxiv.

7. Odd Covers of Graphs. (with C. Buchanan, A. Clifton, E. Culver, J. O'Neill, P. Rombach, and M. Yin)
Journal of Graph Theory (2023). arxiv.

6. On Turán numbers for disconnected hypergraphs. (with R. Mulas)
Acta Mathematica Hungarica 170, 168–182 (2023). arxiv.

5. On Asymptotic Packing of Geometric Graphs. (with D. Cranston, J. Verstraëte, and A. Wesolek)
Discrete Applied Mathematics 322 (2022). arxiv.

4. Maximal independent sets in clique-free graphs. (with X. He, and S. Spiro)
European Journal of Combinatorics (2022). arxiv.

3. Ramsey Numbers for Non-trivial Berge Cycles. (with J. Verstraëte)
SIAM Journal on Discrete Mathematics (2022). arxiv.

2. Triangle-free Subgraphs of Hypergraphs. (with S. Spiro, and J. Verstraëte)
Graphs and Combinatorics (2021). arxiv.

1. *Randomized greedy algorithm for independent sets in regular uniform hypergraphs with large girth. (with J. Verstraëte)
Random Structures & Algorithms (2021). arxiv.

Teaching

Instructor, Georgia Institute of Technology

MATH 1553: Introduction to Linear Algebra	Summer 2026
MATH 1554: Linear Algebra	Summer 2026
MATH 3012: Applied Combinatorics	Fall 2024, 2026
MATH 3406: A Second Course on Linear Algebra	Fall 2025
MATH 4032: Combinatorial Analysis	Spring 2025, 2026

Instructor, Fudan University

Graph Theory	Fall 2023

Instructor, UC San Diego

MATH 20C: Calculus and Analytic Geometry for Science and Engineering	Summer 2021

Teaching Assistant, UC San Diego

Linear Algebra; Calculus; Vector Calculus; Introduction to Analysis;
Discrete Mathematics and Graph Theory; Numerical Analysis; Combinatorics; Complex Analysis.

Service

Co-orgarnizing the GaTech Combinatorics Seminar
Co-orgarnizing the SCMS Combinatorics Seminar
Served as referee for the following journals:

- Journal of London Mathematical Society
- Forum of Mathematics Sigma
- Journal of Combinatorial Theory Series B
- Combinatorica
- Random Structures & Algorithms
- Combinatorics, Probability and Computing
- SIAM Journal of Discrete Mathematics
- Journal of Graph Theory
- European Journal of combinatorics
- Discrete Applied Mathematics
- Discrete Mathematics
- Electronic Journal of combinatorics
- Graphs and Combinatorics
- Australasian Journal of Combinatorics
- Journal of Combinatorial Optimization